

Interpretable Statistical Modeling for Real-Time Sepsis-3 Onset Prediction in MIMIC-IV: A Dynamic Landmarking Approach with Explicit Assumption Testing and Informative-Missingness Analysis

Abdul Ahad

Department of Computer Science & Applied Physics
Atlantic Technological University (ATU)

Galway, Ireland
ahadarif.1998@gmail.com

Abstract—Sepsis is a leading cause of intensive-care mortality, and early recognition improves survival. Recent prediction research is dominated by machine learning and, on the MIMIC-IV database specifically, by *post-diagnosis* prognostic modeling (mortality nomograms) rather than *pre-onset* early detection. The one statistical real-time onset benchmark, TREWScore, was built on the older MIMIC-II database in 2015 and has not been revisited on MIMIC-IV with purely statistical, interpretable tools. We develop and internally validate a fully interpretable statistical pipeline for real-time Sepsis-3 onset prediction on MIMIC-IV (65,241 adult ICU stays), reaching continuously updated risk through a dynamic landmark supermodel rather than any neural or ensemble method. Three methodological commitments distinguish the work. First, the proportional-hazards assumption underlying Cox regression is *formally tested* (Schoenfeld residuals) and, because it fails globally, a time-varying reformulation is fitted in response, a step routinely omitted in the MIMIC-IV Cox literature. Second, incident (new-onset) sepsis is isolated from prevalent sepsis using an hour-6 landmark, exposing that standard bedside scores are non-discriminative to *inverse* for this task (SOFA AUROC 0.36, qSOFA 0.49, SIRS 0.59), whereas a multivariable landmark model attains AUROC 0.775. Third, an informative-missingness analysis shows that *whether a laboratory test was ordered* is a stronger predictor than its value, quantifying a label-recognition ceiling intrinsic to retrospectively labeled ICU data. The pipeline is closed-form, transportable, and pre-wired for external validation on eICU-CRD, directly targeting the internal/external validation asymmetry that recent systematic reviews identify as the field's central unresolved issue. All extraction, modeling, and evaluation code is released for reproducibility.

Index Terms—sepsis prediction, survival analysis, landmarking, proportional hazards, informative missingness, MIMIC-IV, interpretable models, electronic health records

I. INTRODUCTION

Sepsis, defined by the Sepsis-3 consensus as life-threatening organ dysfunction caused by a dysregulated host response to infection [1], remains a major driver of ICU mortality worldwide. Because early treatment materially improves outcomes, predicting onset ahead of clinical recognition is of direct clinical value. Two modeling traditions address this

problem. The first, and by far the more actively published, is machine learning; the second, older and still foundational, is classical statistics, namely Cox proportional-hazards regression, logistic regression, and trajectory modeling. Statistical models retain a specific advantage for real-time clinical use: they yield directly interpretable hazard and odds ratios, they can be checked against formal assumptions, and they can be benchmarked transparently against established rule-based scores (SIRS, qSOFA, NEWS).

On the MIMIC-IV database [2], however, the statistical literature has developed unevenly. A companion review of eighteen peer-reviewed statistical studies found the body of work methodologically mature but concentrated on the wrong half of the problem: most studies predict mortality *after* sepsis is diagnosed rather than onset itself, the proportional-hazards assumption is rarely tested before a Cox model is fitted, and external validation is largely absent from the onset-prediction literature specifically. The single statistical real-time onset benchmark, the TREWScore of Henry et al. [3], was fitted on the older MIMIC-II database in 2015; Liaw et al. [4] used an independent prospective cohort. Neither used MIMIC-IV.

This paper develops a statistical pipeline that addresses these limitations within a single framework and, critically, states its *research gap*, *research question*, and *contributions* explicitly (Sections III–V) so that the novelty relative to prior work is auditable. We deliberately restrict the modeling to classical statistics, both for interpretability and to establish a rigorous, transportable baseline against which machine-learning approaches can later be compared.

II. RELATED WORK

A. Statistical onset prediction

TREWScore [3] fitted a Cox model to routinely collected MIMIC-II vitals and labs to anticipate septic shock, reaching AUC 0.83 with a median 28-hour lead time, the foundational statistical benchmark for real-time sepsis prediction. Liaw

et al. [4] proposed a longitudinal, time-varying Cox extension tracking evolving mortality risk from repeatedly measured indicators. Kasal et al. [5] compared a standard Cox model against Gray’s flexible survival model in severe sepsis and found that hazards are frequently *non-proportional*, a methodological caution largely ignored by later MIMIC-IV work.

B. Prognostic modeling on MIMIC-IV

A recurring MIMIC-IV pattern couples LASSO variable selection with multivariable logistic regression, presented as a nomogram, to predict mortality after diagnosis; several such studies include external validation on eICU-CRD [6]. Trajectory approaches such as group-based trajectory modeling (GBTM) of SOFA scores [7] model the evolving course but again target mortality, not onset. This sub-literature is well validated but prognostic, not predictive of onset.

C. Benchmarks, validation, and reviews

The PhysioNet/CinC 2019 Challenge [8] established a shared early-prediction benchmark and a clinical-utility metric. Fleuren et al. [9] reviewed machine-learning sepsis prediction and reported large between-study heterogeneity. Most pointedly, Wang et al. [10] reviewed 91 sepsis real-time prediction models and found that median AUROC falls from 0.886 internally to 0.783 under full external validation, evidence that the internal/external gap, not raw internal discrimination, is the field’s central weakness.

D. Landmarking and informative missingness

Dynamic prediction by landmarking [11] and its supermodel formulation [12] pool multiple prediction windows to emit continuously updated risk from a single interpretable model, providing a statistical route to real-time prediction without recurrent networks. Separately, informative missingness (the fact that *whether* a test is ordered encodes clinician suspicion) is well documented in EHR modeling [13], [14]. Both threads are central to our design and, to our knowledge, have not been combined for statistical Sepsis-3 onset prediction on MIMIC-IV.

III. RESEARCH GAP

Synthesizing Section II, three gaps are specific and actionable:

- **G1 (Onset vs. prognosis).** MIMIC-IV statistical work overwhelmingly predicts mortality *after* diagnosis. Genuine pre-onset, real-time detection with purely statistical tools has not been revisited on MIMIC-IV since TREWScore’s 2015 MIMIC-II study.
- **G2 (Untested proportional hazards).** Kasal et al. [5] showed sepsis hazards are frequently non-proportional, yet no reviewed MIMIC-IV Cox study reports testing the assumption, weakening confidence in the reported hazard ratios.
- **G3 (Validation asymmetry).** External validation is common in the prognostic nomogram literature but essentially

absent from onset-prediction and trajectory studies, precisely where reviews show performance degrades most [10].

IV. RESEARCH QUESTION AND OBJECTIVES

These gaps motivate one primary research question and three sub-questions.

RQ. *Can a purely statistical (Cox proportional-hazards / time-varying / landmark) model, using routinely available real-time vitals and labs in MIMIC-IV, predict impending Sepsis-3 onset ahead of clinical recognition, with performance comparable to or exceeding the rule-based scores SIRS, qSOFA, and NEWS?*

- **SQ1.** Which routinely measured variables are the strongest statistically significant predictors of onset, and does the proportional-hazards assumption hold for the fitted Cox model?
- **SQ2.** Does modeling trajectories through a time-varying / landmark extension improve early detection over a static single-time-point model?
- **SQ3.** How does discriminative performance compare with SIRS, qSOFA, and NEWS, and does it survive the transition from internal to external (eICU-CRD) evaluation?

The null hypothesis for SQ1 is that proportional hazards holds for at least half of the included covariates; its rejection promotes the time-varying reformulation (Objective under G2) from a secondary to a primary contribution.

V. CONTRIBUTIONS

This work makes the following contributions, each mapped to the gap it closes.

- C1. First interpretable statistical real-time Sepsis-3 onset model on MIMIC-IV (closes G1).** We reach continuously updated, hourly risk through a dynamic landmark supermodel [11], [12], a classical-statistics alternative to recurrent or boosted models, and explicitly separate *incident* from *prevalent* sepsis using an hour-6 landmark, since ~34% of septic stays are already septic at admission and would otherwise inflate apparent performance.
- C2. Explicit proportional-hazards assumption testing with a triggered time-varying reformulation (closes G2).** We report the Schoenfeld residual test for the sepsis-onset Cox model; the global test *fails*, and a time-varying Cox model is fitted in direct response. This diagnostic step is routinely omitted in the MIMIC-IV Cox literature, so reporting it, and acting on it, is itself a methodological contribution.
- C3. Reframing the task: bedside severity scores do not predict incident onset.** On the correct incident-onset target, SOFA is *inversely* discriminative (AUROC 0.36), qSOFA is near chance (0.49), and SIRS is weak (0.59), while a multivariable landmark model reaches 0.775. This quantifies a distinction the field routinely blurs: severity scores detect *established* organ failure, not impending onset.
- C4. Informative-missingness characterization and a label-recognition ceiling.** We show that binary “was

this lab ordered” indicators are stronger predictors than the imputed values themselves, connecting the result to the “curse of knowing” [13], [14] and quantifying an upper bound on what *any* model can achieve when the onset label is itself gated by clinician recognition. This is an honest, reusable characterization result, not a leaderboard claim.

C5. A reproducible, transportable pipeline pre-wired for external validation (targets G3). Because the nomogram and supermodel are closed-form, their coefficients transport directly to a second database; the pipeline is structured so that eICU-CRD [15] validation requires only re-running the cohort/onset/landmark stages, directly addressing the validation asymmetry that reviews flag as decisive [10].

VI. DATA AND METHODS

A. Cohort and labeling

From MIMIC-IV v3.1 we build an adult ICU cohort (age ≥ 18 , first ICU stay only, length of stay ≥ 4 h), yielding 65,241 stays. Suspected infection follows Seymour et al. [16]: antibiotic orders paired with cultures within the Sepsis-3 windows, with surveillance swabs (e.g. MRSA screens) excluded to avoid inflating the suspicion rate. Sepsis-3 is suspected infection together with an acute SOFA rise ≥ 2 ; prevalence is 37.1%, with septic shock 9.2%.

B. Hourly panel and onset labeling

Every stay is resampled to an hourly grid (~ 3.1 M stay-hours). For each variable we retain the raw value, a forward-filled value (what the clinician currently knows), and a time-since-last-measured counter. Hourly SOFA locates the exact onset hour and defines a per-hour “onset within 12h” label. An hour-6 landmark yields 48,829 at-risk stays, of which 3,330 develop onset within 12 h.

C. Models

Baselines are the rule-based scores SIRS, qSOFA, and NEWS computed directly from vitals. Statistical models comprise: (i) a static Cox model with Schoenfeld testing and a time-varying reformulation on failure; (ii) an elastic-net penalized logistic *nomogram* at the hour-6 landmark using an informative-missingness design (median impute each lab, then add a “was measured” indicator); (iii) a *landmark supermodel* stacked across landmarks at 6, 12, 18, 24, 36, and 48 h, so one model emits hourly-updating risk; (iv) GEE with cluster-robust standard errors for within-patient correlation; (v) GBTM of first-24h SOFA trajectories linked to onset and mortality; and (vi) confounder-adjusted causal estimates (PSM, IPW, augmented IPW, doubly robust) for early antibiotics. All randomness uses a fixed seed for reproducibility.

D. Evaluation

Discrimination uses time-dependent AUROC; calibration uses Brier score and calibration curves; clinical utility uses

TABLE I
DISCRIMINATION AT THE HOUR-6 LANDMARK (INCIDENT ONSET WITHIN 12 H).

Model / Score	AUROC
SOFA at hour 6	0.36 (inverse)
qSOFA at hour 6	0.49
SIRS at hour 6	0.59
Multivariable logistic + trajectory slopes	0.755
Full-feature elastic-net nomogram (C1)	0.775

TABLE II
REAL-TIME MODEL AND SUPPORTING ANALYSES (INTERNAL MIMIC-IV).

Component	Headline result
Nomogram	AUROC 0.775; “was-measured” indicators dominate (C4)
Supermodel	Hourly-updating risk; AUROC(s) 0.65–0.77 (C1)
GEE / TV-Cox	Within-patient corr. 0.68; cluster-robust SE up to $4.5\times$ naive
Utility	Normalized utility 0.29; 90% sensitivity $\Rightarrow$ 6.5 false alarms/patient-day
Subgroups	Calibration ratio 0.94–1.06 across sex/age
GBTM	3 latent SOFA-trajectory classes; steepest-worsening $\sim 5\times$ mortality HR
Causal	Early antibiotics (≤ 3 h) associated with lower mortality across PSM/IPW/AIPW/DR

the PhysioNet/CinC 2019 utility score [8] and an alarm-burden curve (false alarms per patient-day at fixed sensitivity). Subgroup calibration across sex and age serves as an internal fairness check and as a stand-in until eICU-CRD external validation is executed.

VII. RESULTS

A. Benchmarking on the incident-onset task

Table I reports discrimination at the hour-6 landmark for incident onset within 12 h. The rule-based scores are non-discriminative to inverse; the statistical landmark models clearly exceed them.

The trajectory-slope terms add statistically significant signal over the static model (DeLong $p \approx 3 \times 10^{-7}$), supporting SQ2.

B. Assumption testing (SQ1)

The global Schoenfeld test rejects proportional hazards for the static onset Cox model, so the null of SQ1 is rejected and a time-varying Cox model is fitted in response, satisfying C2. This confirms, on MIMIC-IV, the non-proportionality that Kasal et al. [5] reported elsewhere.

C. Real-time behavior and honest ceilings

Table II summarizes the deployable model and its supporting analyses. Two results are deliberately framed as *characterizations* rather than wins: the strongest nomogram predictors are missingness indicators (C4), and the PhysioNet utility is modest because the onset label is gated by clinician recognition, an intrinsic ceiling rather than a model deficiency.

VIII. DISCUSSION

The central empirical finding is that the ceiling on incident-onset prediction is set by the label's dependence on clinician recognition (when cultures and antibiotics are ordered) rather than by model class. This is why the strongest predictors are missingness indicators and why a purely statistical model is competitive with far more complex approaches: no neural network is required to approach a ceiling imposed by the labeling process itself. The result aligns with, and sharpens, the informative-missingness literature [13], [14] by placing it at the center of a real-time onset task rather than treating it as a nuisance.

Methodologically, reporting and acting on the proportional-hazards test (C2) restores a diagnostic step the MIMIC-IV Cox literature skips. Reframing the benchmark (C3) corrects a common conflation of severity scoring with onset prediction: SOFA's inverse discrimination is not an anomaly but a direct consequence of scoring *established* dysfunction.

A. Threats to validity

The onset label depends on documentation timing; the assumed-zero SOFA baseline may slightly overcount organ dysfunction; qSOFA uses mean arterial pressure as a proxy for systolic pressure; and the time-varying Cox and HMM analyses use sampled subsets for tractability. Most importantly, all results are *internal* to MIMIC-IV. Given that reviews report a substantial internal-to-external AUROC drop [10], the external validation in C5 is required before any clinical-generalizability claim; it is designed but not yet executed, and is the primary item of future work.

IX. CONCLUSION AND FUTURE WORK

We presented an interpretable, purely statistical pipeline for real-time Sepsis-3 onset prediction on MIMIC-IV that (C1) reaches hourly-updating risk via a landmark supermodel, (C2) tests and acts on the proportional-hazards assumption, (C3) shows bedside scores fail on the incident-onset task, (C4) characterizes an informative-missingness ceiling, and (C5) is transportable and pre-wired for external validation. The immediate next step is eICU-CRD external validation [15] to quantify the internal/external gap for this statistical model; subsequent work will use the same landmark substrate to benchmark machine-learning approaches against this interpretable baseline on identical splits.

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